

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a63f0200d20ccff7f.jsonl`  
- SHA-256: `6a2840c4e3e43e05e1d0e770f2cd542b411f02d625427da051f66ad62ed4253e`  
- Source modified: `2026-06-16T03:24:24+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:22:43.748Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"An ablation on THUMOS14 isolates the contribution of each component: a convolutional baseline pyramid scores 62.1% avg mAP; replacing it with the SGP layer raises this to 68.3% (+6.2), and adding the Trident-head on top yields 69.3% (a further +1.0 over the instant-level regression head) ? quantifying that the relative-boundary head adds a measurable, separable boundary-localization gain."

****Source:**** https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf (primary)

****Supporting quote:**** "compared with the baseline model we implement (Row 1), the SGP layer brings a 6.2% absolute improvement in the average mAP... we compare our Trident-head with the normal instant-level regression head"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:22:46.767Z)

I'll verify this claim about the TriDet paper's THUMOS14 ablation. Let me fetch the primary source.

3. assistant (2026-06-16T03:22:50.933Z)

Let me fetch the TriDet paper to verify the specific ablation numbers claimed.

4. user (2026-06-16T03:23:02.771Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. assistant (2026-06-16T03:23:04.978Z)

The CVF site blocked the fetch. Let me try the arXiv version and search for the specific numbers.

6. user (2026-06-16T03:23:10.501Z)

I cannot locate the ablation study table with those specific mAP numbers in the provided content. The webpage excerpt shows only the abstract and metadata for the TriDet paper, which mentions achieving "69.3%" average mAP on THUMOS14 but does not include the detailed ablation study breakdown you're asking about.

To find the exact baseline, SGP, and Trident-head component contributions, you would need to access the full paper PDF, which is available at the link provided but not included in this webpage excerpt.

7. user (2026-06-16T03:23:14.740Z)

Web search results for query: "TriDet "Relative Boundary Modeling" CVPR 2023 ablation SGP layer Trident-head THUMOS14 average mAP 62.1 68.3 69.3"

Links: [{"title": "TriDet - CVPR 2023 Open Access Repository", "url": "https://openaccess.thecvf.com/content/CVPR2023/html/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.html"}, {"title": "TriDet/README.md at master · dingfengshi/TriDet", "url": "https://github.com/dingfengshi/TriDet/blob/master/README.md"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?", "url": "https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf"}, {"title": "GitHub - dingfengshi/TriDet: [CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://github.com/dingfengshi/TriDet"}, {"title": "CVPR Poster TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://cvpr.thecvf.com/virtual/2023/poster/21956"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/10203543/"}, {"title": "[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/abs/2303.07347"}, {"title": "[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | Request PDF", "url": "https://www.researchgate.net/publication/373307731_TriDet_Temporal_Action_Detection_with_Relative_Boundary_Modeling"}]

I found relevant information about TriDet from your search query. Here are the key findings:

TriDet is a one-stage framework for temporal action detection that proposes a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary.

In the feature pyramid of TriDet, a Scalable-Granularity Perception (SGP) layer is proposed to

aggregate information across different temporal granularities.

Regarding the specific performance metrics you mentioned, TriDet achieves an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%.

The paper was published at CVPR 2023, pages 18857-18866, with authors including Dingfeng Shi, Yujie Zhong, Qiong Cao, Lin Ma, Jia Li, and Dacheng Tao.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:23:18.067Z)

The search confirms 69.3% is the final reported average mAP on THUMOS14. Let me get the full ablation table from the arXiv HTML version or PDF to verify the intermediate numbers (62.1 baseline, 68.3 with SGP).

9. user (2026-06-16T03:23:25.052Z)

I cannot find the ablation study table you're referring to in the provided PDF content. The document appears to be a scientific paper with embedded images and compressed data, but the specific THUMOS14 ablation study table with the exact mAP values (62.1, 68.3, 69.3) you mentioned is not readable in the text extraction provided.

To locate this information, you would need to:

1. Access the PDF directly and navigate to Section 6.2 or the ablation studies section
2. Look for a table comparing baseline, baseline + SGP layer, and + Trident-head results
3. Check the text surrounding the table for the "6.2%" improvement reference

If you can share the readable text from that specific section, I'd be happy to help analyze the numerical comparisons and provide the exact quotes with proper formatting.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579910495-xf7r2b.pdf]

10. assistant (2026-06-16T03:23:27.882Z)

The PDF was saved locally. Let me read the ablation section directly from the saved PDF.

11. user (2026-06-16T03:23:29.065Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

12. assistant (2026-06-16T03:23:31.473Z)

PDF rendering isn't available. Let me extract the text directly using Python's pdftotext or a PDF library.

13. user (2026-06-16T03:23:36.387Z)

```
=== KW '62.1' count 1
=== KW '68.3' count 8
=== KW '69.3' count 9
=== KW '6.2' count 10
=== KW 'instant-level' count 5
=== KW 'SGP layer' count 12
=== KW 'Trident-head' count 33
=== KW 'regression head' count 5
=== KW 'baseline' count 14
```

14. assistant (2026-06-16T03:23:39.603Z)

All the key numbers are present in the PDF. Let me extract the actual ablation context to verify the specific structure of the claim.

15. user (2026-06-16T03:23:43.136Z)

--- CONTEXT 62.1 ---

et.

Main components analysis. We demonstrate the effectiveness of our proposed components in TriDet: SGP layer and Trident-head. To verify the effectiveness of our SGP layer, we use a baseline feature pyramid used by [21, 49] to replace our SGP layer. The baseline consists of two 1D-convolutional layers and shortcut. The window size of convolutional layers is set to 3 and the number of channels of the intermediate feature is set to the same dimension as the Table 5. Analysis of the Effectiveness of three main components on THUMOS14.

Method SA SGP Trident 0.3 0.5 0.7 Avg.

1 77.3 65.2 40.0 62.1

2 ? 82.1 71.0 43.9 66.8

3 ? 83.6 71.7 45.8 68.3

4 ? ? 83.6 72.9 47.4 69.3

Table 6. Analysis of computation cost on THUMOS14. Main: All parts of the model except the detection head. *: Our method with a normal instant-level regression head.

Method mAP GMACs Latency

(ms)0.3 0.7 Avg. Main Head All

ActionFormer82.1 43.9 66.8 30.8 14.4 45.3 224

TriDet? 83.6 45.8 68.3 14.5 14.4 28.9 145

TriDet 83.6

16. assistant (2026-06-16T03:23:45.852Z)

I now have the exact ablation table (Table 5). Let me examine this carefully against the claim, because the structure matters. Let me also get the SA column meaning and the 6.2% / instant-level context.

17. user (2026-06-16T03:23:50.149Z)

=== 6.2 contexts ===

... as our framework. As depicted in Tab. 5, compared with the baseline model we implement (Row 1), the SGP layer brings a 6.2% absolute improvement in the average mAP. Secondly, we compare the SGP with the previous state-of-the-art method...

... size is 19). This change brings a dramatic performance increase in average mAP compared with the CNN baseline (about 6.2%) but is still behind the Transformer baseline by about 1.9%. Next, we conduct experiments with different normal...

18. assistant (2026-06-16T03:23:54.075Z)

Now I have the critical detail. Let me carefully parse what the "+6.2%" refers to. The paper says "the SGP layer brings a 6.2% absolute improvement." Let me check the table structure: Row 1 = 62.1 (baseline, no SA/SGP/Trident), Row 3 = 68.3 (SGP). $68.3 - 62.1 = 6.2$. So the SGP improvement of +6.2 is measured from Row 1 (62.1) to Row 3 (68.3). Let me verify the instant-level regression head comparison and the SA column.

19. user (2026-06-16T03:23:57.274Z)

=== instant-level contexts ===

...e major difference between the Transformer layer and SGP layer is the replacement of the self-attention module with the fully-convolutional module SGP. The successful Layer Normalization [3] (LN) is changed to Group Normalization [43] (GN). As shown in Fig. 4, SGP contains two main branches: an instant-level branch and a window-level branch. In the ?SGP GNLN

FFNx ? x AvgPool FC + ReLU FC Convw Convkw Convw ? ? ?? Instant-level Window-level Shortcut
?Self- attention LNLN FFNx ? Transformer Layer SGP Layer ?? ?? ?????? ???????? Figure 4.
Illustration of the structure of SGP layer. We replace...

... ? ? ?? Instant-level Window-level Shortcut ?Self- attention LNLN FFNx ? Transformer Layer
SGP Layer ?? ?? ?????? ???????? Figure 4. Illustration of the structure of SGP layer. We replace
the self-attention and the second Layer Normalization (LN) with SGP and Group Normalization
(GN), respectively. instant-level branch, we aim to increase the feature discrim- inability
between action and non-action instant by enlarging their feature distance with the video-level
average feature. The window-level branch is designed to introduce the se- mantic content from a
wider receptive field with a branch ?...

...ry predicts the boundary accord- ing to the global feature of the predicted instance seg-
ments. They only consider global information instead of detailed information at each instant.
The second cate- gory directly predicts the distance between an instant and its corresponding
boundary based on the instant-level fea- ture [21, 34, 49, 51]. Some of them refine the segment
with boundary features [21, 34, 51]. However, they do not take the relation (i.e. relative
probability of being a boundary) of adjacent instants into account. The proposed Trident-head
differs from these two categories and sho...

... main components on THUMOS14. Method SA SGP Trident 0.3 0.5 0.7 Avg. 1 77.3 65.2 40.0 62.1 2
? 82.1 71.0 43.9 66.8 3 ? 83.6 71.7 45.8 68.3 4 ? ? 83.6 72.9 47.4 69.3 Table 6. Analysis of
computation cost on THUMOS14. Main: All parts of the model except the detection head. *: Our
method with a normal instant-level regression head. Method mAP GMACs Latency (ms)0.3 0.7 Avg.
Main Head All ActionFormer82.1 43.9 66.8 30.8 14.4 45.3 224 TriDet? 83.6 45.8 68.3 14.5 14.4
28.9 145 TriDet 83.6 47.4 69.3 14.5 29.1 43.7 167 intermediate dimension in the FFN in our SGP
layer. All other hyperparameters (e.g. ...

...elf-attention mechanism in a sliding window behavior [4] with window size 7 (Row 2). We can
see our SGP layer still has 1.5% improvement in average mAP, demonstrating that the
convolutional network can also have excellent performance in TAD task. Besides, we compare our
Trident-head with the normal instant-level regression head, which regresses the boundary
distance for each instant. We can see that the Trident-head improves the average mAP by 1.0%,
and the mAP improvement is more obvious in the case of high IoU threshold (e.g. 1.6% average mAP
improvement in IoU 0.7). Computational complexity...

20. assistant (2026-06-16T03:24:13.958Z)

Now I have the complete picture. Let me carefully evaluate the claim against the actual Table 5 and surrounding text.

****Table 5 (Analysis of three main components on THUMOS14):****

Row	SA	SGP	Trident	0.3	0.5	0.7	Avg.
1				77.3	65.2	40.0	*62.1*
2	?			82.1	71.0	43.9	66.8
3		?		83.6	71.7	45.8	*68.3*
4		?	?	83.6	72.9	47.4	*69.3*

Now let me evaluate the claim's components precisely:

- **Claim 1:**** "a convolutional baseline pyramid scores 62.1% avg mAP" ? Row 1 = 62.1. The text says "The baseline consists of two 1D-convolutional layers and shortcut." ? Accurate.
- **Claim 2:**** "replacing it with the SGP layer raises this to 68.3 (+6.2)" ? Row 3 = 68.3. Paper text: "compared with the baseline model we implement (Row 1), the SGP layer brings a 6.2% absolute improvement in the average mAP." 68.3 ? 62.1 = 6.2. ? Accurate. (Note: Row 2 is the SA/self-attention/Transformer baseline = 66.8, not part of this comparison; the claim correctly skips it.)
- **Claim 3:**** "adding the Trident-head on top yields 69.3 (a further +1.0 over the instant-level regression head)" ? Row 4 = 69.3. Paper text: "we compare our Trident-head with the normal instant-level regression head... the Trident-head improves the average mAP by 1.0%." 69.3 ? 68.3 = 1.0. ? Accurate. The "+1.0 over the instant-level regression head" is exactly what the paper

states. The Trident-head row (Row 4, SGP+Trident=69.3) vs the SGP-only-with-instant-level-regression-head (Row 3 = 68.3, which is also TriDet* in Table 6) ? +1.0. ?

****Claim 4:**** "quantifying that the relative-boundary head adds a measurable, separable boundary-localization gain" ? This is the claim's interpretive framing. The paper confirms the Trident-head models the action boundary via relative probability distribution, and the +1.0 gain is "more obvious in the case of high IoU threshold (e.g. 1.6% average mAP improvement in IoU 0.7)" ? directly supporting that it's a boundary-localization gain that is separable and measurable. ? Accurate interpretation.

Every numerical assertion in the claim matches the primary source exactly. The supporting quote is genuine and present in the paper. Source quality is the primary peer-reviewed CVPR 2023 paper ? appropriate for the claim strength.

One minor nuance to check skeptically: the claim phrases it as a clean sequential pyramid "62.1 ? 68.3 ? 69.3," skipping Row 2 (66.8, the SA/Transformer baseline). This is a legitimate selection ? Rows 1, 3, 4 form the exact ablation path the paper itself uses for the SGP and Trident attributions (the paper explicitly compares Row 3 vs Row 1 for SGP's +6.2, and Row 4 vs Row 3 for Trident's +1.0). The claim does not misrepresent Row 2; it correctly isolates the convolutional baseline (Row 1) rather than the transformer baseline (Row 2). No overreach.

I attempted to refute on three fronts and all failed: (1) the numbers are exact, not misread; (2) no contradicting source ? the paper itself is the source and self-consistent (Table 6 independently confirms 68.3 for TriDet* and 69.3 for TriDet); (3) source is the primary peer-reviewed paper, current (CVPR 2023), not marketing.

21. user (2026-06-16T03:24:20.954Z)

Structured output provided successfully

22. assistant (2026-06-16T03:24:24.421Z)

Verification complete. The claim is ****not refuted**** ? every numerical assertion matches TriDet's Table 5 exactly (Row 1=62.1 conv baseline, Row 3=68.3 with SGP for +6.2, Row 4=69.3 with Trident-head for +1.0 over the instant-level regression head), confirmed verbatim against the primary CVPR 2023 paper and cross-checked by Table 6. High confidence.